Linear Algebra

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1 Vector Spaces

Definition 1.1. A *vector space* over a field $\mathbb{F}$ is a set V with an addition and a scalar multiplication such that, $\forall u, v, w \in V$ and $\forall a, b \in \mathbb{F}$ we have

(1) $u + w = w + u$;
(2) $(u + w) + v = u + (w + v)$;
(3) $\exists 0 \in V$ such that $v + 0 = v$;
(4) $\exists -v$ for all v such that $v + (-v) = 0$;
(5) $1v = v$ and $a(bv) = (ab)v$;
(6) $a(u + v) = au + av$ and $(a + b)u = au + bv$

Notation. For all $w, v \in V$ we have $w + (-v) := w - v$.

Notation. V denotes a vector space over $\mathbb{F}$, unless otherwise stated.

Example 1.1. $\mathbb{R}^n$ is a vector space over $\mathbb{R}$.

Example 1.2. $\mathbb{C}^n$ is a vector space over $\mathbb{C}$ and $\mathbb{R}$.

Example 1.3. $\mathbb{F}^n$ is a vector space over $\mathbb{F}$.

Notation. $\mathbb{F}^\infty = \{(x_1, x_2, \ldots) \mid x_i \in \mathbb{F} \text{ for } k = 1, 2, \ldots\}$

Notation. Let $\mathbb{F}^S$ be the set of functions from $S \to F$. Then,

(1) For $f, g \in \mathbb{F}^S$, the *sum* $f + g \in \mathbb{F}^S$ is defined as
$$(f + g)(x) = f(x) + g(x) \quad \forall x \in V$$
(2) For $f \in \mathbb{F}^S$, the *scalar multiplication* $\lambda f \in \mathbb{F}^S$ is defined as
$$(\lambda f)(x) = \lambda \cdot f(x) \quad \forall x \in V$$

Example 1.4. $\mathbb{F}^S$ is a vector space over $\mathbb{F}$.

Theorem 1.1. *A vector space has a unique additive identity.*

Proof. Suppose 0 and $0'$ are both additive inverses of a vector space V. Then,
$$0 = 0 + 0' = 0'$$
∎

Theorem 1.2. *Every vector space has a unique additive inverse.*

Proof. Suppose w_1 and w_2 are additive inverses of v. Then,

$$w_1 = w_1 + 0 = w_1 + (v + w_2) = (w_1 + v) + w_2 = 0 + w_2 = w_2$$

■

Lemma 1.1. *For all $v \in V$, we have $0v = 0$.*

Proof. $0v = (0+0)v = 0v + 0v \Rightarrow 0v + 0v - 0v = 0v - 0v \Rightarrow 0v = 0.$ ■

Lemma 1.2. *For every $a \in \mathbb{F}$ we have $a0 = 0$*

Proof. $a0 = a(0+0) = a0 + a0 \Rightarrow a0 + a0 - a0 = a0 - a0 \Rightarrow a0 = 0.$ ■

Lemma 1.3. *For all $v \in V$, we have $(-1)v = -v$.*

Proof. We know that,

$$0 = 0v = (1-1)v = 1v + (-1)v = v + (-1)v \Rightarrow 0 - v = v + (-1)v - v$$
$$\Rightarrow -v = (-1)v$$

■

Definition 1.2 (Subspace)**.** A subset U of V is called a *subspace* of V if U is also a vector space with the same additive identity, addition, and scalar multiplication as on V.

Theorem 1.3. *A subset U of V is a subspace of V if and only if U satisfies the following three condition.*

(1) $0 \in U$
(2) $u, w \in U$ *then* $u + w \in U$
(3) $u \in U$ *then* $\lambda u \in U$ *for all* $\lambda \in \mathbb{F}$.

Proof. Suppose $(\Rightarrow)$ U is a subspace then all of the things mentioned above automatically follow.

Suppose $(\Leftarrow)$ U is a subset that satisfies those 3 condition. Then we can quickly check the vector axioms to verify U is indeed a vector space. Thus, U is a subspace of V. ■

Example 1.5. The set $\{(a, b, 5c, c) \in \mathbb{F}^4 \mid a, b, c \in \mathbb{F}\}$ is a subspace of $\mathbb{F}^4$.

Definition 1.3 (sum of vector spaces)**.** Suppose $V_1, \ldots, V_m$ are subspaces of V. The *sum* of $V_1, \ldots, V_m$ denoted by $V_1 + \cdots + V_m$ is defined as

$$V_1 + \cdots + V_m := \{v_1 + \cdots + v_m \mid v_1 \in V_1, \ldots, v_m \in V_m\}$$

Example 1.6. If $U = \{(x, 0) \in \mathbb{F}^2 \mid x \in \mathbb{F}\}$ and $W = \{(0, y) \in \mathbb{F}^2 \mid y \in \mathbb{F}\}$ then

$$U + W = \{(x, y) \in \mathbb{F}^2 \mid x, y \in \mathbb{F}\}$$

Theorem 1.4. *Suppose $V_1, \ldots, V_m$ are the subspaces of V. Then $V_1 + \cdots + V_m$ is the smallest subspace of V containing $V_1, \ldots, V_m$.*

Proof. It is easy to see that $V_1 + \cdots + V_m$ is a vector space. We prove it is the smallest subspace of V containing $V_1, \ldots, V_m$ by arguing that if there exists any other vector space W, containing $V_1, \ldots, V_m$ then $V_1 + \cdots + V_m \subseteq W$.

Let $v_1 \in V_1, \ldots, v_m \in V_m$. Since W contains $V_1, \ldots, V_m$ it must contain $v_1, \ldots, v_m$ hence their sum. Thus, $V_1 + \ldots + V_m \in W$. ■

Definition 1.4 (direct sum)**.** Suppose $V_1, \ldots, V_m$ is a subspace of V. Then the sum $V_1 + V_2 + \cdots + V_m$ is called *direct sum* if and only if each element of the sum can be written in only one way as a sum $v_1 + \cdots + v_m$ where each $v_k \in V_k$.

NOTATION. If $V_1 + \cdots + V_m$ is a direct sum then we write it as $V_1 \oplus \cdots \oplus V_m$.

Example 1.7. Let $V_k = \{(0, \ldots, \underbrace{x_k}_{k^{th}\text{slot}}, \ldots) \in \mathbb{F}^n \mid x_k \in \mathbb{F}\}$. Then, V_k is a subspace of $\mathbb{F}^n$.

Also,

$$\mathbb{F}^n = V_1 \oplus \cdots \oplus V_n$$

Theorem 1.5. *Suppose $V_1, \ldots, V_m$ are subspaces of V. Then $V_1 \oplus \cdots \oplus V_m$ if and only if the only way to write 0 as a sum $v_1 + \ldots + v_m$, where each $v_k \in V_k$, is by taking each $v_k = 0$.*

Proof. Suppose $V_1 + \ldots + V_k$ is a direct sum. Then, 0 can be only written in one way that is

$$0 = \underbrace{0 + 0 + \cdots + 0}_{n}$$

Suppose the only way to write 0 is by adding other 0's. Assume on the contrary that $V_1 + \cdots + V_m$ is not a direct sum. Then, $\exists v$ such that it can be written in two different ways,

$$v = v_1 + \cdots + v_n = u_1 + \cdots + u_n$$
$$\implies (v_1 - u_1) + (v_2 - u_2) + \cdots + (v_n - u_n) = 0$$

By our assumption that 0 can only be written in one way, we have $v_i - u_i = 0 \Rightarrow v_i = u_i$ for $1 \leq i \leq n$. ∎

Theorem 1.6. *Suppose U and W are subspaces of V. Then,*

$$U \oplus W \iff U \cap W = \{0\}$$

Proof. Suppose $U \oplus W$ and $U \cap W \neq \{0\}$. Take a non zero $z \in U \cap W$. But

$$z = z + 0 = 0 + z$$

which violates the assumption $U + W$ is a direct sum. Thus, $\nexists z \in U \cap W$ such that $z = 0$. Since, $0 \in W$ and $0 \in U$ we have $U \cap W = \{0\}$.

Suppose $U \cap W = \{0\}$ and $U + W$ is not a direct sum. Since it is not a direct sum, let us write $0 = u + w$ for some $u \in U, w \in W$. But that implies $u = -w$ which implies $u = w = 0$ due to our assumption. Thus, the only way to write out 0 is as sum of other 0s, which violates Theorem 1.5. ∎

1.1 Problems

EXERCISE 1.1. For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$.

(a) $X_1 = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$
(b) $X_2 = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
(c) $X_3 = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 x_2 x_3 = 0\}$
(d) $X_4 = \{(x_1, x_2, x_3) \in \mathbf{F}^3 : x_1 = 5x_3\}$

Proof. We need to check 3 things.

1) whether $(0, 0, 0)$ is in the subspace.
2) whether it is closed under vector addition.
3) whether it is closed under scalar multiplication.

Now, we do this for each of the four sets.

(a) We know $(0,0,0) \in X_1$ as $0+2\cdot 0+3\cdot 0=0$.
Also, if (x_1,x_2,x_3) and $(y_1,y_2,y_3) \in X_1$ then

$$\begin{aligned}&(x_1+y_1)+2(x_2+y_2)+3(x_3+y_3)\\&=(x_1+2x_2+3x_3)+(y_1+2y_2+3y_3)\\&=0+0\\&=0\end{aligned}$$

Also, if $(x_1,x_2,x_3) \in X_1$ and $c \in \mathbb{F}$ then for $c(x_1,x_2,x_3)$ we have

$$\begin{aligned}cx_1+2cx_2+3cx_3&=c(x_1+2x_2+3x_3)\\&=c\cdot 0\\&=0\end{aligned}$$

So, X_1 is a subspace.

(b) Here, $(0,0,0) \notin X_2$ as $0+2\cdot 0+3\cdot 0=0\neq 4$. So, X_2 is not a subspace.

(c) Take $(1,1,0)$ and $(0,0,1)$. Both are in X_3 as $1\cdot 1\cdot 0=0$ and $0\cdot 0\cdot 1=0$. But

$$(1,1,0)+(0,0,1)=(1,1,1)\notin X_3$$

as $1\cdot 1\cdot 1=1\neq 0$. So, X_3 is not closed under vector addition and hence it is not a subspace.

(d) Clearly, $(0,0,0) \in X_4$ as $0=5\cdot 0$.
Also, if (x_1,x_2,x_3) and $(y_1,y_2,y_3) \in X_4$ then $x_1=5x_3$ and $y_1=5y_3$. Thus,

$$x_1+y_1=5x_3+5y_3=5(x_3+y_3)$$

so $(x_1+y_1,x_2+y_2,x_3+y_3) \in X_4$.
Also, if $(x_1,x_2,x_3) \in X_4$ and $c \in \mathbb{F}$ then $cx_1=c(5x_3)=5(cx_3)$, so $c(x_1,x_2,x_3) \in X_4$.
So, X_4 is a subspace.

Thus, X_1 and X_4 are subspaces of $\mathbb{F}^3$ while X_2 and X_3 are not. ■

EXERCISE 1.2. Verify all assertions given below.

(a) If $b \in \mathbf{F}$, then

$$X=\{(x_1,x_2,x_3,x_4)\in \mathbf{F}^4 : x_3=5x_4+b\}$$

is a subspace of $\mathbf{F}^4$ if and only if $b=0$.

(b) The set of continuous real-valued functions on the interval $[0,1]$ is a subspace of $\mathbf{R}^{[0,1]}$.

(c) The set of differentiable real-valued functions on $\mathbf{R}$ is a subspace of $\mathbf{R}^{\mathbf{R}}$.

(d) The set of differentiable real-valued functions f on the interval $(0,3)$ such that $f'(2)=b$ is a subspace of $\mathbf{R}^{(0,3)}$ if and only if $b=0$.

(e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbf{C}^{\infty}$.

Proof.

(a) Suppose 'X' is a subspace then $(0,0,0,0)$ is in X. So,

$$0=5\cdot 0+b \implies b=0$$

Now, suppose $b=0$ then $X=\{(x_1,x_2,x_3,x_4)\in \mathbb{F}^4 : x_3=5x_4\}$. Here, $(0,0,0,0)\in X$ as $0=5\cdot 0$. Also, if (x_1,x_2,x_3,x_4) and $(y_1,y_2,y_3,y_4)\in X$ then

$$x_3+y_3=5x_4+5y_4=5(x_4+y_4)$$

so their sum is in X. Also, if $c\in\mathbb{F}$ then $cx_3=c(5x_4)=5(cx_4)$, so $c(x_1,x_2,x_3,x_4)\in X$. So, X is a subspace if and only if $b=0$.

(b) Let X be the set of continuous real-valued functions on $[0,1]$. It is because the zero function $0(x) = 0$ is continuous (it is constant), the sum of two continuous functions is continuous and if f is continuous then cf is also continuous. To see this, take $f, g \in X$, $c \in \mathbb{R}$ and $a \in [0,1]$. Then,

$$\lim_{x\to a}(f+g)(x) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x) = f(a) + g(a) = (f+g)(a)$$

and

$$\lim_{x\to a}(cf)(x) = c\lim_{x\to a} f(x) = cf(a) = (cf)(a)$$

So, $f+g, cf \in X$ and hence X is a subspace of $\mathbb{R}^{[0,1]}$.

(c) Let X be the set of differentiable real-valued functions on $\mathbb{R}$. Similar reasoning as above. The zero function is differentiable with $0' = 0$. Also, if $f, g \in X$ and $c \in \mathbb{R}$ then $f+g$ and cf are differentiable and

$$(f+g)' = f' + g' \quad \text{and} \quad (cf)' = cf'$$

So, X is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) Let $X = \{f : (0,3) \to \mathbb{R} \mid f'(2) = b \text{ and } f \text{ is differentiable}\}$ then, we need to prove X is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

Suppose it is a subspace then, $0 \in X$. Thus, $b = 0'(2) = 0$ (as $0(x) = 0$ is a constant function).

Suppose $b = 0$ then, $0 \in X$ as 0 is differentiable and $0'(2) = 0$. Also,

$$f'(2) = 0 \text{ and } g'(2) = 0 \implies (f+g)'(2) = f'(2) + g'(2) = 0$$

Also, $f'(2) = 0 \implies (cf)'(2) = cf'(2) = c \cdot 0 = 0$. Here, $f+g$ and cf are differentiable as well. So, X is a subspace.

(e) We know, $\mathbb{C}^\infty = \{(x_1, x_2, \ldots) \mid x_i \in \mathbb{C}\}$. Now, let

$$L = \{(x_n)_{n=1}^\infty \in \mathbb{C}^\infty \mid \lim_{n\to\infty} x_n = 0\}$$

Then, $(0,0,\ldots) \in L$. Also if $(x_n)_{n=1}^\infty \in L$ and $(z_n)_{n=1}^\infty \in L$ then,

$$\begin{aligned}\lim_{n\to\infty}(x_n + z_n) &= \lim_{n\to\infty} x_n + \lim_{n\to\infty} z_n \\ &= 0 + 0 = 0\end{aligned}$$

Now, if $(x_n)_{n=1}^\infty \in L$ and $c \in \mathbb{C}$ then,

$$\lim_{n\to\infty} cx_n = c\lim_{n\to\infty} x_n = c \cdot 0 = 0$$

So, L is a subspace of $\mathbb{C}^\infty$.

■

EXERCISE 1.3. Show that the set of differentiable real-valued functions f on the interval $(-4,4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.

Proof. Let $X = \{f : (-4,4) \to \mathbb{R} \mid f'(-1) = 3f(2) \text{ and } f \text{ is diff}^n\}$

Now, we know that $0 \in X$ as $0(x) = 0$ is diff^n and $0'(-1) = 0 = 3 \cdot 0(2)$.

Now, suppose f and $g \in X$ then $f+g$ is diff^n and,

$$\begin{aligned}(f+g)'(-1) &= f'(-1) + g'(-1) \\ &= 3f(2) + 3g(2) \\ &= 3(f+g)(2)\end{aligned}$$

Also, if $f \in X$ and $c \in \mathbb{R}$ then cf is diff^n and, $(cf)'(-1) = cf'(-1) = c \cdot 3f(2) = 3(cf)(2)$

So, X is indeed a subspace of $\mathbb{R}^{(-4,4)}$. ■

EXERCISE 1.4. Suppose $b \in \mathbf{R}$. Show that the set of continuous real-valued functions f on the interval $[0,1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbf{R}^{[0,1]}$ if and only if $b = 0$.

Proof. Let $X = \{f : [0,1] \to \mathbb{R} \mid \int_0^1 f = b \text{ and } f \text{ is cont}^n\}$ then, suppose X is a subspace then $0 \in X$

Then, $b = \int_0^1 0 = 0$.

Now, suppose $b = 0$ then, it is immediately clear that $0(x) = 0$ is in X as $\int_0^1 0 = 0$ and 0 is continuous. Also, if $f, g \in X$ then $f + g$ is continuous and

$$\int_0^1 (f+g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0$$

And, if $c \in \mathbb{R}$ then cf is continuous and

$$\int_0^1 cf = \int_0^1 c \cdot f = c \int_0^1 f = c \cdot 0 = 0$$

So, it is indeed a subspace. ■

EXERCISE 1.5. Is $\mathbf{R}^2$ a subspace of the complex vector space $\mathbf{C}^2$?

Proof. No. Here, $\mathbb{C}^2$ is a complex vector space so the scalars are from $\mathbb{C}$. Now, $(0,1) \in \mathbb{R}^2$ but

$$i(0,1) = (0,i) \notin \mathbb{R}^2$$

as $i \notin \mathbb{R}$. So, $\mathbb{R}^2$ is not closed under scalar multiplication and hence it is not a subspace of $\mathbb{C}^2$.

Note, if we consider $\mathbb{C}^2$ as a vector space over $\mathbb{R}$ then $\mathbb{R}^2$ is a subspace, as $(0,0) \in \mathbb{R}^2$ and it is closed under addition and under multiplication by real scalars. ■

EXERCISE 1.6. (a) Is $\{(a,b,c) \in \mathbf{R}^3 : a^3 = b^3\}$ a subspace of $\mathbf{R}^3$?
(b) Is $\{(a,b,c) \in \mathbf{C}^3 : a^3 = b^3\}$ a subspace of $\mathbf{C}^3$?

Proof. For (a) let $X = \{(a,b,c) \in \mathbb{R}^3 \mid a^3 = b^3\}$ then we know,

$$\begin{aligned}(a-b)(a^2+ab+b^2) &= 0 \\ \implies a = b \text{ or } a^2+ab+b^2 &= 0.\end{aligned}$$

Now, if $a^2 + ab + b^2 = 0$ then

$$\left(a + \frac{b}{2}\right)^2 + \frac{3}{4}b^2 = 0 \implies b = 0 \text{ and } a = 0$$

as both the terms are non-negative. So, in either case $a = b$. Also, if $a = b$ then clearly $a^3 = b^3$. Thus,

$$X = \{(a,a,c) \in \mathbb{R}^3\}$$

Now, it is a subspace because $(0,0,0) \in X$

Also, $(a,a,c) + (b,b,d) = (a+b, a+b, c+d) \in X$

and, $k(a,a,c) = (ka, ka, kc) \in X$.

So, X is a subspace of $\mathbb{R}^3$.

Now, for (b) let $X = \{(a,b,c) \in \mathbb{C}^3 \mid a^3 = b^3\}$

Here, $\omega = \frac{-1+i\sqrt{3}}{2} \in \mathbb{C}$ so that $1 + \omega + \omega^2 = 0$ and $\omega^3 - 1 = (\omega - 1)(\omega^2 + \omega + 1) = 0$, i.e. $\omega^3 = 1$.

Then, $(\omega, 1, 0)$ and $(1, 1, 0) \in X$ because

$$\omega^3 = 1 = 1^3 \text{ and } 1^3 = 1^3$$

But $(\omega, 1, 0) + (1, 1, 0) = (\omega + 1, 2, 0) \notin X$ as,

$$\begin{aligned}(\omega + 1)^3 &= \omega^3 + 1^3 + 3\omega(\omega + 1) = 2 + 3(\omega^2 + \omega) \\ &= -1 \neq 2^3 = 8\end{aligned}$$

So, X is not closed under addition and hence it is not a subspace of $\mathbb{C}^3$. ■

EXERCISE 1.7. Prove or give a counterexample: If U is a nonempty subset of $\mathbf{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbf{R}^2$.

Proof. The statement is false. For now, consider $U = \{(a, -a) \mid a \in \mathbb{Z}\}$

Clearly, $U \subseteq \mathbb{R}^2$ and U is nonempty as $(0, 0) \in U$. Also, if $(a, -a) \in U$ then $-(a, -a) = (-a, a) \in U$ as $-a \in \mathbb{Z}$. And,

$$(a, -a) + (b, -b) = (a + b, -(a + b)) \in U$$

as $a + b \in \mathbb{Z}$. But, $c(a, -a) = (ca, -ca)$ need not be in U. For eg: $(1, -1) \in U$ but

$$\tfrac{1}{2}(1, -1) = \left(\tfrac{1}{2}, -\tfrac{1}{2}\right) \notin U$$

as $\frac{1}{2} \notin \mathbb{Z}$. So, U is not closed under scalar multiplication and hence it is not a subspace of $\mathbb{R}^2$. ■

EXERCISE 1.8. Give an example of a nonempty subset U of $\mathbf{R}^2$ such that U is closed under scalar multiplication, but U is not a subspace of $\mathbf{R}^2$.

Proof. Take the set,

$$U = \{(a, b) \in \mathbb{R}^2 \mid |a| = |b|\}$$

Here,

$$(0, 0) \in U, \text{ and if } (a, b) \in U \text{ and } c \in \mathbb{R} \text{ then } |ca| = |c||a| = |c||b| = |cb|$$

so, $(ca, cb) \in U$. But, U is not closed under addition. For eg: $(1, 1), (-2, 2) \in U$ but

$$(1, 1) + (-2, 2) = (-1, 3) \notin U$$

as $|-1| = 1 \neq 3 = |3|$. So, U is not a subspace of $\mathbb{R}^2$. ■

EXERCISE 1.9. A function $f\colon \mathbf{R} \to \mathbf{R}$ is called *periodic* if there exists a positive number p such that $f(x) = f(x + p)$ for all $x \in \mathbf{R}$. Is the set of periodic functions from $\mathbf{R}$ to $\mathbf{R}$ a subspace of $\mathbf{R}^{\mathbf{R}}$? Explain.

Proof. No. Let P be the set of periodic functions from $\mathbb{R}$ to $\mathbb{R}$. Take the functions,

$$f(x) = x \bmod \pi, \quad g(x) = x \bmod 1$$

Here, $x \bmod t = x - t\lfloor x/t \rfloor$ for $t > 0$, so $0 \leq x \bmod t < t$. Now,

$$f(x + \pi) = x + \pi - \pi \left\lfloor \frac{x}{\pi} + 1 \right\rfloor = x + \pi - \pi \left(\left\lfloor \frac{x}{\pi} \right\rfloor + 1 \right) = f(x)$$

and similarly $g(x + 1) = g(x)$, so $f, g \in P$.

We claim $f + g \notin P$. Suppose it is, then there is $p > 0$ s.t $(f + g)(x + p) = (f + g)(x)$ for all $x \in \mathbb{R}$. Now, $(f + g)(0) = 0$ so

$$f(p) + g(p) = (f + g)(p) = (f + g)(0) = 0$$

But $f(p) \geq 0$ and $g(p) \geq 0$, so $f(p) = g(p) = 0$. Thus, $p = \pi\lfloor p/\pi \rfloor$ and $p = \lfloor p \rfloor$, i.e. $p = k\pi$ and $p = m$ for some positive integers k, m (as $p > 0$). So,

$$\pi = \frac{m}{k} \in \mathbb{Q}$$

which is a contradiction as π is irrational. So, $f + g \notin P$.

Note, $0 \in P$ and P is closed under scalar multiplication (a period of f is also a period of cf), so it is only the addition that fails. Thus, P is not a subspace of $\mathbb{R}^{\mathbb{R}}$. ■

EXERCISE 1.10. Suppose V_1 and V_2 are subspaces of V. Prove that the intersection $V_1 \cap V_2$ is a subspace of V.

Proof. Since $0 \in V_1$ and $0 \in V_2$ we have $0 \in V_1 \cap V_2$.

Also, if $a, b \in V_1 \cap V_2$ then $a + b \in V_1$ and $a + b \in V_2$ as both are closed under addition, so $a + b \in V_1 \cap V_2$.

Also, if $a \in V_1 \cap V_2$ and $c \in \mathbb{F}$ then $ca \in V_1$ and $ca \in V_2$ as both are closed under scalar multiplication, so $ca \in V_1 \cap V_2$.

So, $V_1 \cap V_2$ is a subspace of V. ■

EXERCISE 1.11. Prove that the union of two subspaces of V is a subspace of V if and only if one of the subspaces is contained in the other.

Proof. Suppose V_1 and V_2 are subspaces of V. Then, let $V_1 \cup V_2$ be a subspace of V. Suppose $V_1 \not\subseteq V_2$ and $V_2 \not\subseteq V_1$. Then, $\exists a \in V_2$ s.t $a \notin V_1$ and $\exists b \in V_1$ s.t $b \notin V_2$. Now, $a, b \in V_1 \cup V_2$ so $a + b \in V_1 \cup V_2$ so either $a + b \in V_1$ or $a + b \in V_2$. If $a + b \in V_1$, then $(a + b - b) = a \in V_1$ which is a contradiction as $a \notin V_1$ and if $a + b \in V_2$ then, $(a + b - a) = b \in V_2$ which is a contradiction as $b \notin V_2$. Both of them lead to a contradiction. So, $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$.

Now, suppose $V_1 \subseteq V_2$ s.t V_1 and V_2 are subspaces. Then, $V_1 \cup V_2 = V_2$ which is a subspace. Similarly, if $V_2 \subseteq V_1$ then $V_1 \cup V_2 = V_1$ which is a subspace. ■

EXERCISE 1.12. Prove that the union of three subspaces of V is a subspace of V if and only if one of the subspaces contains the other two.

Proof. Suppose V_1, V_2, V_3 are subspaces of V. First, if one of them contains the other two, say $V_3 \supseteq V_1 \cup V_2$ then $V_1 \cup V_2 \cup V_3 = V_3$ which is a subspace (the other cases are same). That is the trivial direction.

Now, let $V_1 \cup V_2 \cup V_3$ be a subspace of V. We have two cases.

Case 1: One of them is contained in the union of the other two, say $V_3 \subseteq V_1 \cup V_2$ (the other cases are same).

Then, $V_1 \cup V_2 \cup V_3 = V_1 \cup V_2$, so $V_1 \cup V_2$ is a subspace and by 1.11 we have $V_1 \subseteq V_2$ or $V_2 \subseteq V_1$. If $V_1 \subseteq V_2$ then $V_3 \subseteq V_1 \cup V_2 = V_2$, so V_2 contains V_1 and V_3. Similarly, if $V_2 \subseteq V_1$ then V_1 contains V_2 and V_3. So, we are done in this case.

Case 2: None of them is contained in the union of the other two. We show that this case is not possible. Now, $V_3 \not\subseteq V_1 \cup V_2$ and $V_2 \not\subseteq V_1 \cup V_3$, so there are a, b s.t

$$a \in V_3,\ a \notin V_1 \cup V_2$$
$$b \in V_2,\ b \notin V_1 \cup V_3$$

Now, $a, b \in V_1 \cup V_2 \cup V_3$ and this is a subspace, so $a + b$ and $-a + b$ are in $V_1 \cup V_2 \cup V_3$.

Suppose $a + b \in V_2$ then $(a + b - b) = a \in V_2$ which is a contradiction.

Suppose $a+b \in V_3$ then $(a+b-a) = b \in V_3$ which is a contradiction.

Now, $a+b \in V_1$. Also, $-a+b \in V_1$ because if $-a+b \in V_2$ then

$$(-a+b-b) \in V_2 \implies -a \in V_2 \implies a \in V_2$$

which is a contradiction and if $-a+b \in V_3$ then $(-a+b+a) = b \in V_3$ which is a contradiction.

So, $-a+b \in V_1$ and $a+b \in V_1 \implies (-a+b)+(a+b) = 2b \in V_1 \implies b \in V_1$ as $2 \neq 0$ in $\mathbb{F}$. But $b \notin V_1$, a contradiction. So, Case 2 is not possible and we are done. ■

EXERCISE 1.13. Suppose

$$U = \{(x,-x,2x) \in \mathbf{F}^3 : x \in \mathbf{F}\} \quad \text{and} \quad W = \{(x,x,2x) \in \mathbf{F}^3 : x \in \mathbf{F}\}.$$

Describe $U+W$ using symbols, and also give a description of $U+W$ that uses no symbols.

Proof. Here, as $(x,-x,2x)+(y,y,2y) = (x+y,-x+y,2(x+y))$ we have

$$U+W = \{(x+y,-x+y,2(x+y)) \in \mathbb{F}^3 \mid x,y \in \mathbb{F}\}$$

Description,

$U+W$ is the set of all vectors with three coordinates whose 3rd coordinate is twice the 1st coordinate.

Now, let $T = \{(a,b,c) \in \mathbb{F}^3 \mid c = 2a\}$. We show $U+W = T$.

Take $(x+y,-x+y,2(x+y)) \in U+W$, its 3rd coordinate is $2(x+y)$ which is twice the 1st. So, $U+W \subseteq T$.

Now, take $(a,b,2a) \in T$ and let $x = \frac{a-b}{2}$, $y = \frac{a+b}{2}$. Then,

$$x+y = a, \quad -x+y = b, \quad 2(x+y) = 2a$$

so $(a,b,2a) = (x+y,-x+y,2(x+y)) \in U+W$. So, $T \subseteq U+W$.

Thus,

$$U+W \subseteq T \;\&\; T \subseteq U+W \implies U+W = T$$

So, the two descriptions above are the same set. ■

EXERCISE 1.14. Suppose U is a subspace of V. What is $U+U$?

Proof.

$$U+U = \{v_1+v_2 \mid v_1,v_2 \in U\}$$

Now, let $v \in U+U$. Then, $v = u_1+u_2$ for some $u_1,u_2 \in U$. Then, $v \in U$ as U is closed under addition. Also, if $v \in U$ then $v = v+0$ with $v,0 \in U$.

Thus, $v \in U+U$. Since,

$$U+U \subseteq U \;\&\; U \subseteq U+U \implies U = U+U$$

So, $U+U$ is just U itself. ■

EXERCISE 1.15. Is the operation of addition on the subspaces of V commutative? In other words, if U and W are subspaces of V, is $U+W = W+U$?

Proof. Yes. Take $a \in U+W$. Then, $a = m+n$ for $m \in U$ and $n \in W$. Now, $a = n+m$ as addition in V is commutative. Thus, $a \in W+U$. Similarly, take $b \in W+U$. Then, $b = i+j$ for some $i \in W$ and $j \in U$. Now, $b = j+i \in U+W$.

Thus,

$$U+W \subseteq W+U \;\&\; W+U \subseteq U+W \implies W+U = U+W.$$

So, addition of subspaces is commutative. ■

EXERCISE 1.16. Is the operation of addition on the subspaces of V associative? In other words, if V_1, V_2, V_3 are subspaces of V, is

$$(V_1 + V_2) + V_3 = V_1 + (V_2 + V_3)?$$

Proof. Yes. Let $v \in (V_1 + V_2) + V_3$ then

$$v = (v_1 + v_2) + v_3 \text{ for some } v_1 \in V_1, v_2 \in V_2, v_3 \in V_3.$$

$$\implies v = v_1 + (v_2 + v_3) \in V_1 + (V_2 + V_3) \text{ as } v_2 + v_3 \in V_2 + V_3.$$

Thus,

$$(V_1 + V_2) + V_3 \subseteq V_1 + (V_2 + V_3)$$

Similarly, let $w \in V_1 + (V_2 + V_3)$ then $w = v_1 + (v_2 + v_3)$ for some $v_1 \in V_1, v_2 \in V_2, v_3 \in V_3$. Now, $w = (v_1 + v_2) + v_3 \in (V_1 + V_2) + V_3$. Thus,

$$V_1 + (V_2 + V_3) \subseteq (V_1 + V_2) + V_3$$

and so,

$$V_1 + (V_2 + V_3) = (V_1 + V_2) + V_3.$$

So, addition of subspaces is associative. ■

EXERCISE 1.17. Does the operation of addition on the subspaces of V have an additive identity? Which subspaces have additive inverses?

Proof. Yes. Take the zero subspace $\{0\}$. Then, for any subspace U of V,

$$U + \{0\} = \{u + 0 \mid u \in U\} = U$$

and $\{0\} + U = U$ by 1.15. So, $\{0\}$ is an additive identity. Also, it is the only one. If E is also an additive identity then

$$E = \{0\} + E = \{0\}$$

where the first equality is as $\{0\}$ is an additive identity and the second is as E is an additive identity (applied to the subspace $\{0\}$).

Now, suppose U has an additive inverse W, i.e. $U + W = \{0\}$. Then, if $u \in U$ we have $u = u + 0 \in U + W = \{0\}$ as $0 \in W$. So, $U \subseteq \{0\}$ and thus $U = \{0\}$. Also, $\{0\} + \{0\} = \{0\}$ so $\{0\}$ is its own additive inverse.

Thus, the only subspace with an additive inverse is $\{0\}$. ■

EXERCISE 1.18. Prove or give counterexample: If V_1, V_2, U are subspaces of V such that

$$V_1 + U = V_2 + U$$

then $V_1 = V_2$.

Proof. The statement is false. Take $V = \mathbb{F}^2$ and

$$V_1 = \{(x, 0) \mid x \in \mathbb{F}\}, \quad V_2 = \{(0, x) \mid x \in \mathbb{F}\}, \quad U = \{(-y, y) \mid y \in \mathbb{F}\}.$$

One can easily check that all of them are subspaces.

Now, let $(a, b) \in \mathbb{F}^2$. Then

$$(a, b) = (a + b, 0) + (-b, b) \in V_1 + U,$$
$$(a, b) = (0, a + b) + (a, -a) \in V_2 + U,$$

as $(-b, b)$ and $(a, -a)$ lie in U (take $y = b$ and $y = -a$). Since $V_1 + U$ and $V_2 + U$ are subsets of $\mathbb{F}^2$, we get

$$V_1 + U = \mathbb{F}^2 = V_2 + U.$$

But $V_1 \neq V_2$, as $(1,0) \in V_1$ whereas $(1,0) \notin V_2$ (every element of V_2 has first coordinate 0). So, the hypothesis holds and the conclusion fails. ■

EXERCISE 1.19. Suppose

$$U = \{(x,x,y,y) \in \mathbb{F}^4 \mid x,y \in \mathbb{F}\}$$

Find a subspace W of $\mathbb{F}^4$ such that $\mathbb{F}^4 = U \oplus W$.

Proof. Take the set $W = \{(0,a,b,0) \in \mathbb{F}^4 \mid a,b \in \mathbb{F}\}$. Notice that $0 \in W$. Also,

$$(0,a,b,0) + (0,c,d,0) = (0,a+c,b+d,0) \in W;$$

$$k \cdot (0,a,b,0) = (0,ka,kb,0) \in W$$

Thus, the set W is a subspace of $\mathbb{F}^4$. Let $\mathbf{v} = (v_1,v_2,v_3,v_4) \in \mathbb{F}^4$. Then,

$$\mathbf{v} = (v_1,v_1,v_4,v_4) + (0,v_2-v_1,v_3-v_4,0) \in U+W$$

Thus, $\mathbb{F}^4 = U+W$. Now let $\mathbf{v} \in U \cap W$. Since $\mathbf{v} \in U \cap W$, we have $\mathbf{v} \in W$. Thus, $\mathbf{v} = (0,v_2,v_3,0)$, but $\mathbf{v} \in U$ as well, which forces $v_2 = 0$ and $v_3 = 0$. Thus, $\mathbf{v} = (0,0,0,0)$. Therefore, $U+W$ is a direct sum via Theorem 1.6. ■

EXERCISE 1.20. Suppose

$$U = \{(x,y,x+y,x-y,2x) \in \mathbb{F}^5 \mid x,y \in \mathbb{F}\}$$

Find a subspace W of $\mathbb{F}^5$ such that $\mathbb{F}^5 = U \oplus W$.

Proof. Take $W = \{(0,0,a,b,c) \in \mathbb{F}^5 \mid a,b,c \mid \mathbb{F}\}$. Then, it is easy to see that W is a subspace of $\mathbb{F}^5$. Set $a = u-x-y, b = v-x+y, c = w-2x$ where $u,v,w,x,y \in \mathbb{F}$ then,

$$U+W = \{(x,y,x+y+a,x-y+b,2x+c)\} = \{(x,y,u,v,w)\} = \mathbb{F}^5$$

Thus, $\mathbb{F}^5 = U+W$. Now, let $\mathbf{v} = (v_1,v_2,v_3,v_4,v_5) \in \mathbb{F}$ such that $\mathbf{v} \in U \cap W$ then $v_1 = v_2 = 0$ because $\mathbf{v} \in W$. Since $\mathbf{v} \in U$ that implies $v_3 = v_1+v_2 = 0, v_4 = v_1-v_2 = 0$, $v_5 = 2v_1 = 0$. Thus, $U \cap W = \{0\}$. Therefore, $\mathbb{F}^5 = U \oplus W$ via Theorem 1.6. ■

EXERCISE 1.21. Suppose

$$U = \{(x,y,x+y,x-y,2x) \in \mathbb{F}^5 \mid x,y \in \mathbb{F}\}$$

Find subspaces W_1, W_2, W_3 of $\mathbb{F}^5$ such that $\mathbb{F}^5 = U \oplus W_1 \oplus W_2 \oplus W_3$.

Proof. ■